

Project: Li Electrodeposition - Glyme Electrolytes

June 4

Goal: Screen electrolyte **240604-B** for stable Li plating at 30°C .

Experimental Conditions

* **Electrolyte Composition:** 1 M LiTFSI in diglyme : EtOH (4:1 v/v), 20 mL ct+.

* **Additive:** 8% by weight, 5 mol% 12-crown-4.

* **Preparation:** Stirred for 20 min.

* Target Temperature (T): 22.4°C .

* Glovebox Conditions: $\text{H}_2\text{O} < 1 \text{ ppm}$.

Electrochemical Setup

* **Working Electrode:** Glassy Carbon RDE (0.3 cm^2).

* **Counter Electrode (CE):** Li foil.

* **Reference Electrode (Ref):** Ag/AgCl.

Deposition Run 240604-B1

Conditions Applied: -0.45 V vs Ag/AgCl, duration = 90 min, rotation speed (ω) = 1600 rpm.

* Current Density: $j = 0.50 \frac{\text{mA}}{\text{cm}^2}$.

Calculations of Deposited Mass and Charge

$j = 0.50 \frac{\text{mA}}{\text{cm}^2}$, $A_{\text{elec}} = 0.3 \text{ cm}^2$

$I = (0.5 \times 10^{-3} \frac{\text{A}}{\text{cm}^2}) \cdot (0.3 \text{ cm}^2) = 1.5 \times 10^{-4} \text{ A}$

$t_{\text{dep}} = 90 \text{ min} = 5400 \text{ s}$

$Q = I \cdot t = (1.5 \times 10^{-4} \text{ A}) \cdot (5400 \text{ s}) = 0.81 \text{ C}$

Moles of Li Deposited:

Assuming $n=2$ electrons per Li :

$n_{\text{mol}} := \frac{Q}{2F} = \frac{0.81 \text{ C}}{96485 \text{ C/mol}} = 8.4 \times 10^{-6} \text{ mol}$

Mass of Li Deposited:

$m_{\text{Li}} = n_{\text{mol}} \cdot M_{\text{Li}} = (8.4 \times 10^{-6} \text{ mol}) \cdot (6.94 \text{ g/mol})$

Charge Balance:

$e^- : \text{Li}^+ = 5.8 \times 10^{-5} : 1$

Electrode Temperature Test: Hot Plate at 30°C

* **Observation (F):** 1 M solution appears grey + dull upon deposition.

| Time | Temperature ($^{\circ}\text{C}$) |

| :--- | :--- |

| **0 min** | 22.4°C |

| **1 min** | 23.1°C |

| **5 min** | 25.6°C |

| **10 min** | 27.9°C |

| **20 min** | 30.1°C |

| **40 min** | 31.5°C |

| **1 hr** | 32.0°C |

| **1 hr 30 min** | 32.6°C |

XRD Analysis (Main Peak)

* Position (2θ): 2.1° (Low intensity).

* Feature: Shoulder at $2\theta = 4.7^{\circ}$.

Conclusion and Findings

The screening of electrolyte **240604-B** (1 M LiTFSI in diglyme/EtOH with 5 mol% crown ether) under deposition conditions (-0.45 V , 90 min) resulted in a calculated lithium deposit mass derived from the charge passed ($Q = 0.81 \text{ C}$). The stoichiometry indicates an electron-to-lithium ratio of approximately $5.8 \times 10^{-5} : 1$.

Thermal stability testing revealed that while the target temperature was set at 30°C , significant heating occurred during operation; within one hour and thirty minutes, the solution temperature rose to 32.6°C starting from an initial 22.4°C . Visually, the 1 M electrolyte exhibited a grey and dull appearance post-deposition. XRD analysis identified a low-intensity main peak at 2.1° with a shoulder at 4.7° , suggesting potential phase formation or impurity presence that warrants further investigation regarding crystallographic structure stability in this glyme-based system.